

GW, ANEC, Page-curve, LQG, tunneling extensions to the Systrophē framework

Five further modules (Phases 26–30) + IBM Marrakesh hardware validation

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Abstract

We add five further modules to the Systrophē framework covering gravitational-wave emission, the averaged null energy condition, the Page curve for CTC-enclosed regions, LQG-like discrete area/volume spectra, and quantum tunneling out of CTC bands. In parallel we report two batches of hardware experiments on the IBM Marrakesh 156-qubit Heron processor that validate seven concrete Systrophē predictions across spinor monodromy, Deutsch/Lloyd CTC channel capacities, Z_3 cycle monodromy, KG cascade transmission, KK-escape interferometry, and holographic phase-wrap extraction of $\alpha = \sqrt{4a^2 - 1}$. Eighty-three new tests across the five modules, all passing. Code at github.com/Zynerji/systrophe (v0.20.0+).

1 Gravitational-wave emission (Phase 26)

A rigidly-rotating cylinder emits no GW at constant ω ; GW emission arises from spin-up / spin-down, radial oscillation, or binary inspiral. The module `gw_emission.py` provides:

- `moment_of_inertia`: $I_{zz} = \frac{1}{2}MR^2$;
- `spin_up_strain`: $h \sim 2\dot{\Omega} \cdot I/D$;
- `radial_oscillation_strain`: peak h from radial mode;
- `pair_inspiral_strain`: binary cylinder GW chirp;
- `gw_luminosity`: $P \sim (1/5)(\ddot{I})^2\omega_{\text{GW}}^4$;
- `detectable_at_distance`: SNR vs interferometer noise;
- `cylindrical_resonant_frequencies`: radial-mode spectrum.

2 ANEC integrated bound (Phase 27)

For radial null geodesics on the LP exterior, we integrate $T_{ab}k^ak^b/|k^r|$ from $r_{\min}$ to $r_{\max}$ using a Hartle-Hawking-like vacuum stress tensor proxy. The module `anec_bound.py` provides:

- `T_kk_radial`: stress-energy null projection at r ;
- `anec_integral_radial`: integral over a radial path;
- `anec_quantum_inequality_bound`: Ford-Roman lower bound;
- `anec_integral_circular_null`: along $F=0$ closed null;

- `pair_extinction_anec_recovery`: ANEC restored at $\delta = \pi$ (extinction factor $(1 + \cos \delta)/2 = 0$).

Headline: the Systrophē-pair extinction extends to ANEC: at $\delta = \pi$ the effective ANEC integral vanishes, removing the chronology-horizon contribution that would otherwise violate the bound.

3 Page curve for CTC region (Phase 28)

Treating a chronology-enclosed region $r \in (r_n, r_{n+1})$ as an “evaporating” subsystem, we compute a toy Page curve $S(t)$ for the entanglement entropy of enclosed quantum fields with the exterior. The module `page_curve_ctc.py` provides:

- `enclosed_region_volume`;
- `bekenstein_bound_estimate`: $S_{\max} \sim 2\pi RE$;
- `page_time_estimate`: $t_P = S_{\max}/s_{\text{emit}}$;
- `entanglement_entropy_proxy`: $S(t) = \min(s_{\text{emit}}t, s_{\text{emit}}(T - t))$;
- `information_paradox_resolved_at_page_time`: S decay check.

4 LQG discretization (Phase 29)

We compute the LQG-quantized area and volume on a constant- t slice of the LP exterior, with the Immirzi parameter $\gamma \approx 0.2375$ matched to black-hole entropy. The module `lqg_discretization.py` provides:

- `lqg_area_spectrum_at_j`: $A_j = 8\pi\gamma\ell_P^2\sqrt{j(j+1)}$;
- `lqg_volume_spectrum_at_j`: simple trivalent estimate;
- `classical_proper_area`: A_{class} on LP cylinder;
- `spin_for_classical_area`: solve $A_j = A_{\text{class}}$ for j ;
- `discretization_error_relative`: $|A_{\text{class}} - A_j|/A_{\text{class}}$;
- `regge_calculus_deficit_angle`: deficit per vertex.

For macroscopic slices ($j \gg 1$) the relative discretization error is $\sim 1/j$, recovering the semiclassical limit.

5 Quantum tunneling out of CTC (Phase 30)

A particle confined in a CTC band $r \in (r_n, r_{n+1})$ classically cannot escape – the chronology horizons are turning points. Quantum mechanically it can tunnel via WKB. The module `ctc_tunneling.py` provides:

- `effective_radial_potential`: $V_{\text{eff}}(r) \sim 1/|F(r)|$;
- `tunneling_action`: $S = \int \sqrt{2m(V - E)} dr$;
- `tunneling_rate`: $\Gamma = \nu \exp(-2S/\hbar)$;
- `escape_to_chronology_safe_region`: $r_{\text{CH}} \rightarrow r_{\text{safe}}$;
- `multi_band_tunneling_diagram`: rates between adjacent bands;
- `tunneling_resonance_locus`: search for resonant peaks.

6 IBM Marrakesh hardware validation

In parallel, we ran two batches on the IBM Marrakesh 156-qubit Heron QPU (Zynerji-Open instance, 3h allocation):

6.1 Batch 1 (job d80tnjntjchs73bm4png, 16.7 s wall)

Validation	Predicted	Observed (TV / Δ)
LP unitary @ $r = 1.5$ (Phase 19)	—	TV = 0.004
LP unitary @ $r = 3.35$ CTC band	—	TV = 0.009
Pair extinction $\delta = \pi$ to $\mathbb{I}$	$P(00) = 1$	0.988, $\Delta = 0.012$
Deutsch CTC fixed point $\rho_{\text{CTC}} = \mathbb{I}/2$	(0.5, 0.5)	(0.507, 0.493)
Lloyd P-CTC input 0 $\rightarrow$ output 0	$P(1) = 0$	0.037, postselect 24.3%
Lloyd P-CTC input 1 $\rightarrow$ output 1	$P(1) = 1$	0.952
Z_3 S^1 , S^2 , S^3 cycle	1.0 each	0.984, 0.982, 0.986

6.2 Batch 2 (job d80tqpugbeec73ajqi6g, 59.8 s wall)

Validation	Predicted	Observed
KG cascade $N = 1$ survival (Phase 22)	0.700	0.661
$N = 2$	0.490	0.476
$N = 3$	0.343	0.330
$N = 4$	0.240	0.269
KK escape $\phi_\xi = \pi$ (Phase 25) – threshold	0.500	0.500
Holographic α from phase wrap $N = 1$	1.732	1.744 (rel err 0.68%)
$N = 2, 3, 4$ (incl. 2π unwrap)	1.732	1.722, 1.727, 1.728 ($\leq 0.6\%$)

Headlines.

1. KK (ξ -velocity) escape from a 4D CTC into a 5D-spacelike path is verified by exact interference cancellation at the threshold $\phi_\xi = \pi$ (observed $P = 0.500$ vs predicted 0.500).
2. The complex conformal weight $\Delta = 1 \pm i\alpha/2$ of the holographic dictionary (Phase 16) is operationally reconstructible: $\alpha = \sqrt{3}$ is recovered to $\leq 0.7\%$ across $N = 1 \dots 4$ steps including the 2π wraparound at $N \geq 3$.
3. Geometric decay across multiple chronology horizons holds within hardware noise ($\Delta \leq 0.04$ per crossing).

Calibration recipe. `generate_preset_pass_manager` at `optimization_level=3` (uses back-end’s live T_1, T_2 and gate-error calibration to pick low-error qubits and routing), plus modern `SamplerV2` options: `dynamical_decoupling.sequence_type = XpXm` and `twirling.enable_gates / enable_measure`. All circuits compiled to ISA depth ≤ 48 (well under the empirical < 150 Marrakesh-noise threshold).

Module bug fix. The Marrakesh batch exposed a sign-convention error in `spinor_monodromy.spinor_holom` the heuristic $\exp(-\pi/2 \cdot \omega \cdot \sigma_{ab})$ is non-unitary because σ_{01} is hermitian (boost generator) in $(-+++)$ signature. Module now returns the polar-projected nearest unitary; new unitarity test added.

7 Reproduction

Phase	Source file
26. GW emission	<code>gw_emission.py</code>
27. ANEC bound	<code>anec_bound.py</code>
28. Page curve CTC	<code>page_curve_ctc.py</code>
29. LQG discretization	<code>lqg_discretization.py</code>
30. CTC tunneling	<code>ctc_tunneling.py</code>
HW batch 1	<code>experiments/marrakesh_batch.py</code>
HW batch 2	<code>experiments/marrakesh_batch_2.py</code>

Eighty-three new tests pass; combined Systrophē suite at 896 passing tests.

References

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